General Zoology
(Z102)

By
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Course Description

- This course is designed to study the principles and basics of animal taxonomy (identification, description, nomenclature and classification of different kinds of animals),
- and also to enhance the quality of students in understanding the main structure and life cycles of some parasitic animals.

Aims

- This course verifies the following specific aims:

1. To teach you how to recognize the taxonomy of different animal phyla using their common characters.
2. To provide you with knowledge about the normal biodiversity of animal groups in the ecosystem.
3. To help you predict any future impact of ecological disorders on the normal biodiversity.

Objectives

- Upon completion of this course you should be able to:

1. Use the course understandings to write concise research reports describing the methods used and data required as well as place these finding in the context of published knowledge on the subject.
2. Apply research finding in a broader environmental and social context.
3. Distinguish some parasites and know the life cycles goes and also the methods of infection by the infective stages.

Course Format

- There will be several forms of instruction utilized throughout the semester. Your success in this course is directly dependent upon your effort and participation in:

1. Required departmental lecture notes according to the Contents; you can recognize the text required to read prior to each class meeting.
2. Lecture-room discussions which will expand upon the material presented within the text readings. Lectures will be supported by various hand-outs, drawings and data- show presentation.
3. A media review project that will be assigned during the first 10 weeks of semester. This project aims to afford you the opportunity to investigate the course updating. The scope and format of the project will be reviewed and the required completion form will be provided to you.
4. The discussion board for this course is a valuable resource for further topic exploration and clarification.

- You will also be provided with additional supplemental readings throughout the semester to assist in your comprehension of the course content.
- Please familiarize yourself with whiteboard and speak to me in case you have any issue with access or utilization.

Assessment

- Your progress and overall course achievement will be calculated in the following manner.

Measure from	100 total mark
Mid-term Examination	10 %
Final-term Examination	40 %
Oral Examination	10 %
Practical Examination	30 %
Semester Work	10 %
Total mark	100 %

1. INTRODUCTION

1.1. Definition of Systematic Zoology:

- Systematic Zoology or Animal Taxonomy is the branch of Zoology concerned with the identification, description, nomenclature and classification of different kinds of animals.

1.2. Numbers:

- At present, more than one million species of animals are known.

1.3. Diversity:

- They present an interesting diversity of structure, function, habits and mode of life.

1.4. Classification of living organisms:

- The Greek philosopher Aristotle (384 - 322 BC) was among the earliest scientists who attempted classifying living organisms.
- He suggested that animals could be classified according to the presence or absence of 'red blood' into two groups Enaima and Anaima, with and without red blood respectively.
- It was also once suggested that animals may be classified according to the type of environment they live in as terrestrial, aquatic or aerial,
- or according to the type of food they use into flesh-eaters (carnivorous) and plant-eaters (herbivorous).

1. INTRODUCTION

1.4. Classification of living organisms (cont.):

- Gradually, however, emerged the idea of classifying animals according to the morphological similarities between them. This idea was formulated clearly in the 17th century by the English naturalist **John Ray** (1672 - 1 705) whose system can be considered as the **first scientific** approach to annual **classification**.
- Moreover, he was the first to give a precise definition of the **species**, the basic unit for classification of living organisms.
- He defined the **species** as "an assemblage or grouping of animals which are **morphologically similar** and which interbreed freely with one another but they commonly do not inter-breed with other species, and if they do so they produce infertile (sterile) hybrids".

1. INTRODUCTION

1.4. Classification of living organisms (cont.):

- Later came the Swedish naturalist **Linnaeus** (1707 - 1778) who laid down the basis of the system of classification we use nowadays.
- He according to the **morphological** and **anatomical** similarities between them. He also devised the system of "**Binomial nomenclature**" by which each type of organism is given a name composed of two words, the first is the name of the genus, and the second is the name of the species.
- Both is Latin and are written underlined or typed in italics.
- The name of the genus starts with a capital later and that of the species with a small letter.
- The scientific name of man, for example, is *Homo sapiens*, where Homo denotes the generic name while sapiens represents the specific name.

1. INTRODUCTION

1.4. Classification of living organisms (cont.):

- **Nowadays**, the classification of animals is not only based upon morphological and anatomical characteristics but also on **biochemical**, **genetically**, **embryological** and **physiological** features.
- Species having many features in common are placed in the same **genus**, similarly, related genera are grouped in a family, and **families** with certain common characteristics constitute an **order**, and orders in turn are grouped into classes. From these **classes**, the higher taxonomic groupings known as **phyla** are formed.
- Although the relationships between the different phyla may not be obvious, it is always possible to arrange them according to the degree of complexity of structure and function into larger groups known as **subkingdoms** or branches.

1. INTRODUCTION

1.4. Classification of living organisms (cont.):

- Thus, the Animal Kingdom is classified into three subkingdoms:

I. Subkingdom Protozoa:

- This subkingdom includes a single phylum, Phylum Protozoa, the members of which are referred to as unicellular animals (or a cellular) since the body of each is formed of a single protoplasmic mass that performs all life activities.

II. Subkingdom Parazoa:

- Includes a single phylum, Porifera, the members of which are simple multicellular animals commonly known as sponges.
- The body of the animal is formed of many cells, which are not much differentiated and do not form proper tissues.

III. Subkingdom: Metazoa:

- This includes the rest of the animal phyla.
- The body in these animals is composed of many cells, which are well differentiated into distinct types and form definite tissues.
- Divisions of labor and cell specialization are remarkable in these phyla.
- The different phyla of the Metazoa show gradations from simple structural organization to implicated organ systems in the body.

1. INTRODUCTION

1.1. Definition of systematic zoology:

1.2. Numbers:

1.3. Diversity:

1.4. Classification of living organisms:

2. SUBKINGDOM PROTOZOA (PHYLUM PROTOZOA)

2.1. Species numbers:

- The phylum includes more than 50,000 species, which are found in almost all habitats, wherever moisture is present

2.2. Classification:

- Phylum Protozoa is divided into four subphyla on the basis of the locomotory organelles found in the various species.

I. Subphylum Mastigophora, includes protozoa that move by one or more flagella.

II. Subphylum Sarcodina, includes protozoa that move by pseudopodia.

III. Subphylum Sporozoa, includes protozoa winch have no distinct locomotory organelles. They produce resistant spores towards the end of their life cycle, hence the name Sporozoa.

IV. Subphylum Ciliophora includes protozoa that move by cilia.

2.2. Classification:

I. SUBPHYLUM MASTIGOPHORA

1. Movement:

- Mastigophora includes those protozoa that move by one to several flagella.

2. Habitats:

- Many species live in fresh or marine waters, and some live as parasites.

3. Main character:

- This class is of particular interest because its members combine the characters of plant and animals, and is therefore considered as the most primitive of the protozoan classes.

2. SUBKINGDOM PROTOZOA (PHYLUM PROTOZOA)

2.2. Classification:

I. SUBPHYLUM MASTIGOPHORA

4. Classification:

- It is divided into 2 classes (Phytomastigophora and Zoomastigopbora) according to the presence or absence of chromatophores.
- a) Class Phytomastigophora: includes plant-like free-living flagellates that have chromatophores, e. g. *Euglena*.
- b) Class Zoomastigopbora: these are animal-like flagellates with no chromatcphores. The great majority of this group lives as a parasite in larger animals, e. g. *Trypanosoma* and *Leishmania*.

5. EUGLENA SP.:

5.1. Living and habitat:

- Euglena is a free-living flagellate that lives in fresh water ponds, particularly those that are rich in nitrogenous organic matter.
- The presence of large numbers of this flagellate, especially in laboratory cultures, gives the water a greenish tint لون مخضر.

2. SUBKINGDOM PROTOZOA (PHYLUM PROTOZOA)

2.2. Classification:

I. SUBPHYLUM MASTIGOPHORA

5. EUGLENA SP.:

5.2. Morphology:

- Euglena is a small spindle-shaped organism, measuring less than 0.1 mm in length.
- The anterior end is blunt while the posterior end is pointed.
- A thin, firm but elastic covering known as the pellicle covers the body.
- Distinct spiral ridges mark the pellicle.
- At the blunt anterior end of the body, there is a funnel shaped depression, which starts with the mouth or cytostome and leads through a narrow gullet or cytopharynx into a spherical reservoir.
- A long, whip-like flagellum extends outwards from the cytostome.
- It originates from a conspicuous granule known as the basal granule or blepharoplast found at the bottom of the reservoir.
- Another short flagellum extends in the reservoir, originating from a second blepharoplast.

2.2. Classification:

I. SUBPHYLUM MASTIGOPHORA

5. EUGLENA SP.:

5.2. Morphology (cont.):

- On one side of the gullet, there is a conspicuous, red-pigmented spot known as the stigma or eyespot.
- Very close to the reservoir there is a contractile vacuole system that is formed of a large central vacuole surrounded by a number of smaller ones.
- The cytoplasm is differentiated into an outer clear non-granular ectoplasm and an inner granular endoplasm.
- The nucleus is large, oval and usually lies nearer to the pointed end of the body.
- It encloses a distinct central endosome.
- A number of rod or disc-shaped green bodies are found suspended in the cytoplasm.
- These are the chloroplasts that are responsible for the green coloration of the animal.
- Suspended also in the cytoplasm are some living organelles, such as the mitochondria and Golgi apparatus as well as ovoid paramylum granules formed of a carbohydrate allied to starch (Fig. 1).

2. SUBKINGDOM PROTOZOA (PHYLUM PROTOZOA)

2.2. Classification:

I. SUBPHYLUM MASTIGOPHORA

5. EUGLENA SP.:

5.2. Morphology (cont.):

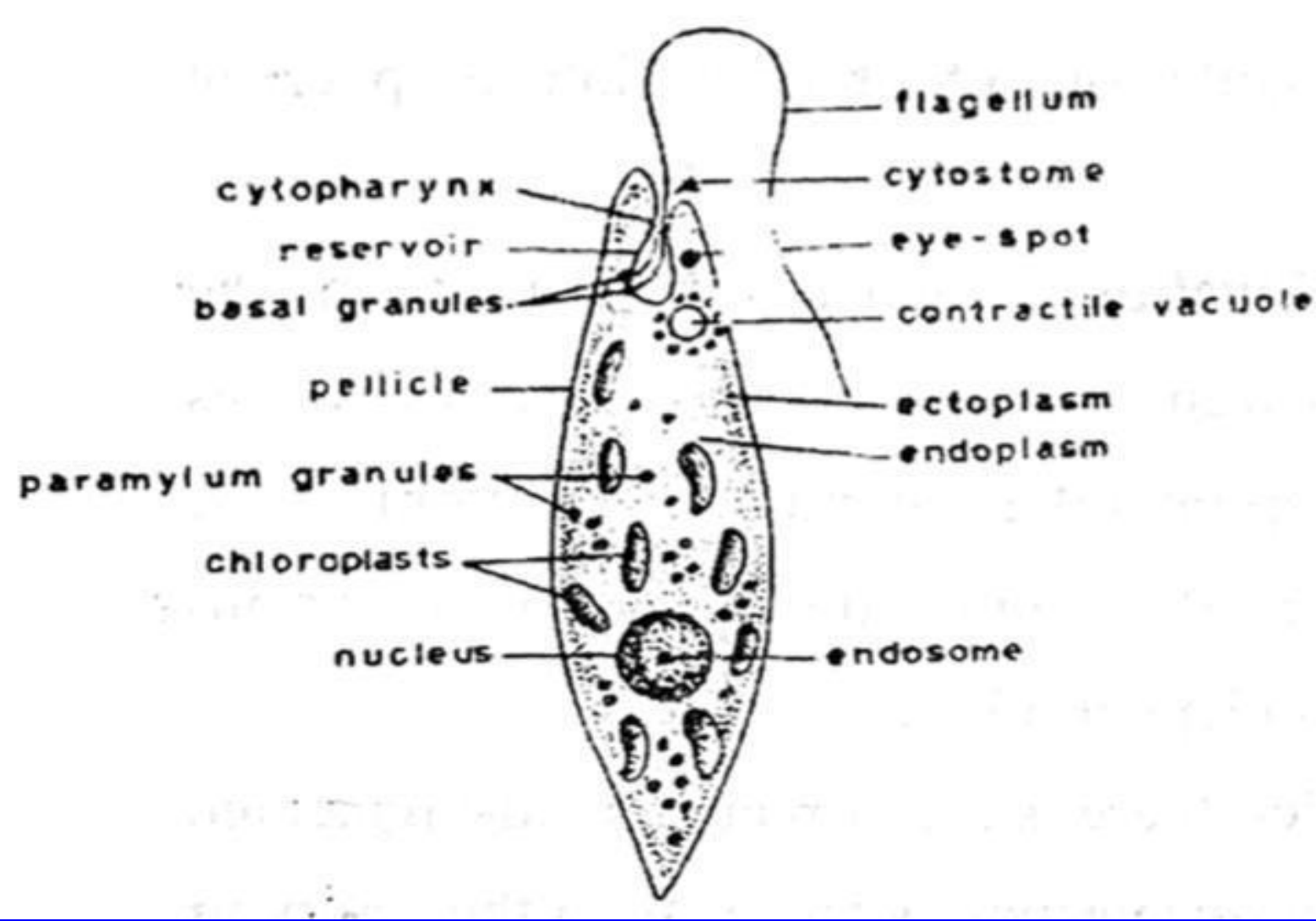

Fig. 1: Morphology of *Euglena sp.*

2.2. Classification:

I. SUBPHYLUM MASTIGOPHORA

5. EUGLENA SP.:

5.3. Locomotion:

- Euglena moves in its environment by **two methods**.
- The **first** is a **slow** worm-like movement known as the **euglenoid movement**. This type of movement is brought about by differential contractions and relaxations of the different parts of the body (Fig. 2).

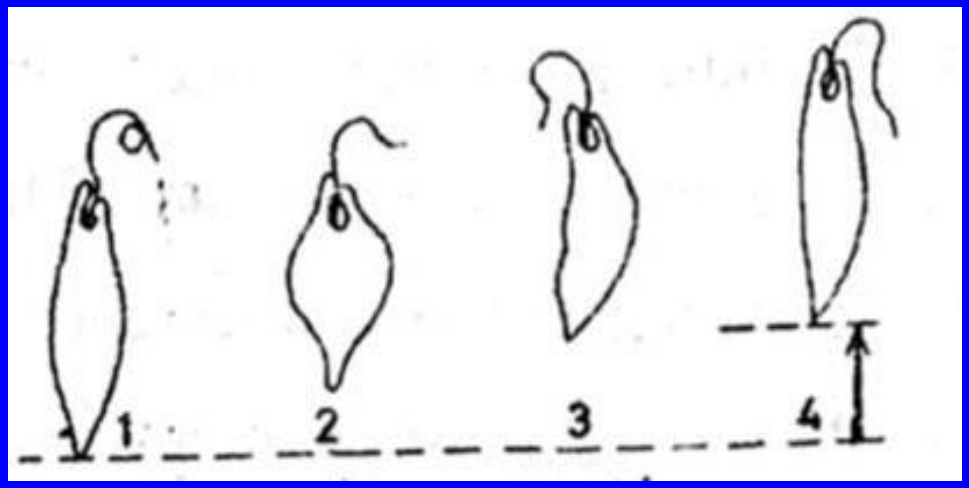

Fig. 2: Euglenoid movement in Euglena sp.

2.2. Classification:

I. SUBPHYLUM MASTIGOPHORA

5. EUGLENA SP.:

5.3. Locomotion (cont.):

- The second type of movement is faster and enables the animal to swim rapidly in water.
- It is carried out by the long flagellum and the mechanical principle involved in this type of movement is the same as that of the screw or propeller.
- The flagellum lashes obliquely backwards with waves passing along the flagellum from base to tip in a spiral manner, with an increase in velocity and amplitude. This results in rotation along the longitudinal axis and whirling about the axis of the general direction of locomotion. The animal then behaves as a rotating inclined plane and thus moves forward through the water (Fig. 3).

2.2. Classification:

I. SUBPHYLUM MASTIGOPHORA

5. EUGLENA SP.:

5.3. Locomotion (cont.):

Fig. 3: Flagellar movement in Euglena sp.

2.2. Classification:

I. SUBPHYLUM MASTIGOPHORA

5. EUGLENA SP.:

5.3. Locomotion (cont.):

- As the animal moves forwards, it also rotates spirally around its longitudinal axis.
- Most probably, this is brought about by the generation of successive undulations along the flagellum during the effective stroke (Fig. 4).

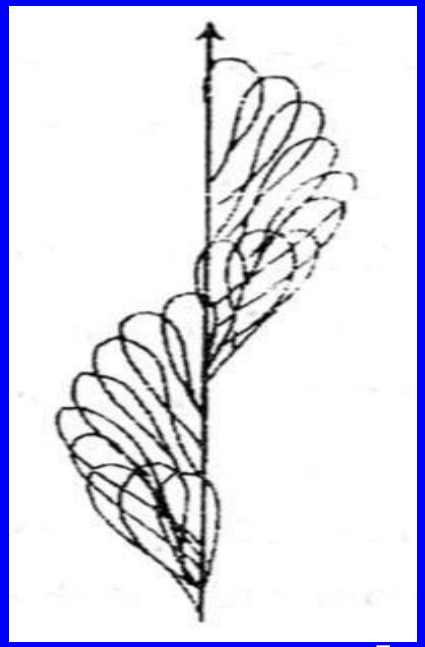

Fig. 4: Spiral movement in *Euglena sp.*

2.2. Classification:

I. SUBPHYLUM MASTIGOPHORA

5. EUGLENA SP.:

5.4. Nutrition:

- In the presence of sunlight, Euglena can synthesize carbohydrates from carbon dioxide and water by the aid of its chloroplasts. This method of nutrition is identical to the process of photosynthesis that occurs in plants and is referred to as holophytic or autotrophic nutrition.
- In the absence of light, certain species of Euglena can absorb soluble food material from the surrounding environment through the body surface by pinocytosis.
- No food is taken in through the cytostome in Euglena and related flagellates.

2.2. Classification:

I. SUBPHYLUM MASTIGOPHORA

5. EUGLENA SP.:

5.5. Osmoregulation:

- Since the outer covering of Euglena is semi-permeable and the concentration of salts is higher in its cytoplasm than in the surrounding fresh water, water tends to flow continuously into its body by osmosis.
- This water should be expelled; otherwise the animal would gradually swell and eventually die.
- Water balance or osmoregulation in Euglena is carried out by the contractile vacuole system.
- Excess water collects as minute droplets in the small vacuoles that empty in the large central vacuole, and this bursts at regular intervals driving this water into the reservoir, from which it passes to the exterior through the cytostome.

2.2. Classification:

I. SUBPHYLUM MASTIGOPHORA

5. EUGLENA SP.:

5.6. Respiration and Excretion:

- Euglena is aerobic and respires the oxygen dissolved in the surrounding water.
- Exchange of oxygen and carbon dioxide takes place by simple diffusion through the body surface.
- Similarly, soluble excretory products of metabolism pass from the cytoplasm to the external environment by simple diffusion.
- During daytime, the process of photosynthesis may mask respiration.
- Carbon dioxide resulting from respiration is utilized in photosynthesis, while the oxygen produced by the latter process is used in respiration.

2.2. Classification:

I. SUBPHYLUM MASTIGOPHORA

5. EUGLENA SP.:

5.7. Reproduction:

- Euglena reproduces only asexually by longitudinal binary fission.
- Immediately before this division starts, the animal stops moving and encloses itself in a thin envelope.
- The nucleus then, divides by mitosis and the various organelles are duplicated.
- Eventually, the animal divide: lengthwise, starting at the anterior end, into two daughter individuals.
- When the prevailing conditions of the environment are unfavorable, the organism is capable of encystment by the secretion of a thick gelatinous cyst around itself.
- Quite frequently, Euglena reproduces by binary fission during encystment.
- In some instances, as many as 16 or 32 young euglenas may be formed in a single cyst.

2.2. Classification:

I. SUBPHYLUM MASTIGOPHORA

5. EUGLENA SP.:

5.8. Sensitivity:

- **Chlorophyll-bearing** euglenas are attracted to ordinary **light**.
- This is expectable particularly when we take into consideration the importance of light in the holophytic nutrition of the organism.
- It responds **negatively** to **strong chemical solutions**, twisting its body and changing its course.

2.2. Classification:

I. SUBPHYLUM MASTIGOPHORA

5. EUGLENA SP.:

5.9. Is Euglena a plant or an animal?

- Some botanists believe that Euglena and related flagellates should be considered as plants rather than animals. This belief is apparently based on the fact that Euglena -as the great majority of plants-possesses chloroplasts and exhibits holophytic nutrition.
- However, most zoologists believe that these organisms should be classified as animals on the following grounds:
- 1- The presence of a contractile vacuole system. Such a system is not known in the plant kingdom.
- 2- None of the known plants move by an organelle comparable in structure to the flagellum.
- 3- The plane of division in binary fission of Euglena and related flagellates is longitudinal while in plants it is transverse.
- 4- The presence of an eyespot in many flagellates and its absence in all known plants.